



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE – RAIGAD – 402 103

Winter Semester Examination – December-2019

Branch: B. Pharm.

Subject: Medicinal Chemistry-I (BP402T)

Date: 06/12/2019

Semester: IV

Marks: 75

Time: 3hrs

- Instructions:** i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Multiple Choice Questions (MCQs)

(20 Marks)

1. Aspirin introduced by -----in 1889
 - A. Dreser's.
 - B. Emil Fischer and Mering.
 - C. Barger and Dale.
 - D. Non of Above.
2. Bioisosteres are similar in their-
 - A. Physical Character.
 - B. Chemical Character.
 - C. Both Character a & b.
 - D. Biochemical Character.
3. Which of the following reaction is not a Phase I metabolic transformation?
 - A. Reduction of ketones.
 - B. Conjugation of alcohols.
 - C. Monoamine oxidases.
 - D. Esterhydrolysis.
4. -----reaction is most common in drugs having ester or amide functional groups
 - A. Oxidative reactions.
 - B. Reduction.
 - C. Hydrolysis.
 - D. Phase I.
5. Which of the following adrenergic drug belongs to category of vasoconstrictors?
 - A. Metaraminol.
 - B. Isoprenaline.
 - C. Amrinone.
 - D. Naphazoline.
6. Which type of Receptor in Bronchial Smooth Muscle.
 - A. Alpha 1 subtype.
 - B. Beta 1 subtype.
 - C. Alpha 2 subtype
 - D. Beta 2 subtype.
7. Prazocin belongs to the class of
 - A. Pyridyl quinoxaline.
 - B. Piprazinyl quinazoline.
 - C. Pyridyl quinazoline
 - D. Piprazinyl quinoxaline
8. In Propranolol the aromatic ring and its substitute is the primary determining of which activity.
 - A. Alpha 1 antagonistic.
 - B. Alpha 2 antagonistic.
 - C. Beta 1 antagonistic.
 - D. Beta 2 antagonistic.
9. Ach agonist drug having IUPAC 2(Acetoxy)-N, N, N-trimethyl propan-1-aminium
 - A. Methacholine.
 - B. Bethanechol.
 - C. Carbachol.
 - D. Pilocarpine.

10. IUPAC name of Pralidoxime chloride is-
 - A. 2-[(Hydroxyamino)methyl]-1-methyl pyridine-1-ium chloride.
 - B. Ethyl-(3-hydroxyhynyl)dimethyl-ammonium chloride.
 - C. 3-[(Dimethylamino carbonyl)oxy]-1-methyl pyridinium chloride.
 - D. 2 (Acetoxy)-N, N, N-trimethyl propan-1-aminium chloride.
11. Tropicamide contain which amine group-
 - A. Primary amine.
 - B. Secondary amine.
 - C. Tertiary amine.
 - D. Contain no amine group.
12. Which replacement is general structure of anti-cholinergic drug make it useful in case of peptide ulcer patient?
 - A. Replacement of N from tertiary to quaternary.
 - B. Change in R1, R2, R3.
 - C. Length of alkyl chain.
 - D. None of the above.
13. One of the following is not a triazolobenzodiazepine derivative
 - A. Alprazolam.
 - B. Triazolam.
 - C. Midazolam.
 - D. Estazolam.
14. Blood levels of neuroleptics are useful to check the following:
 - A. Toxicability.
 - B. Therapeutic levels.
 - C. Resistace.
 - D. All of the above.
15. The Mechanism of action of Carbamazepine appears to be similar to that of:
 - A. Benzodiazepines.
 - B. Sodium valproate.
 - C. Phenytoin.
 - D. Ethosuximide.
16. The anesthetic agent, which is an inhibitor of NMDA glutamate receptors is:
 - A. Thiopental.
 - B. Halothane.
 - C. Ketamine.
 - D. Sevoflurane.
17. Which COX enzyme are aspirin, indomethacin, and sulindac more selective for?
 - A. COX-1.
 - B. COX-2.
 - C. They are equally selective for COX-1 & COX-2.
18. Most current opioid analgesics act at this opioid receptor subtype
 - A. Kappa.
 - B. Delta.
 - C. Epsilon
 - D. Mu.
19. Drug associated with the hepatic/renal toxic metabolite: N-acetyl-p-benzoquinone.
 - A. Diclofenac.
 - B. Meclofenamate.
 - C. Indomethacin.
 - D. Acetaminophen.
20. What can be done to prevent GI side effect of NSAIDs?
 - A. Co-administration of NSAID with PPI.
 - B. Enteric coated tablets.
 - C. Prescribe CINODs instead.
 - D. All of the above.

Q. No.2 Long Answers (Answer any two)

(20 Marks)

- A] Classify Antipsychotics with example & Give SAR of any three categories. Sketch out Synthetic pathway of phenytoin.
- B] Define Drug metabolism. Write about Phase-I & Phase-II reactions.
- C] Give the classification of Narcotic analgesics. Discuss SAR & MOA of Morphine Analogue. Draw schematic pathway for synthesis of methadone.

Q. No.3 Short Answers (Answer any Seven)

(35Marks)

- A] Explain classification of bioisosterism.
- B] Elaborate Factors affecting drug metabolism.
- C] Write few example of mixed action of sympathetic drug with IUPAC & structure.
- D] Highlights two example of Nonselective alpha adrenergic blocker with MOA & Structure.
- E] What are cholinergic receptor? Classify them. Sketch out synthetic pathway of Halothane & Ketamine.
- F] Classify anticonvulsant with example. Discuss SAR of Barbiturates.
- G] Give SAR & MOA of Fenamates.
- H] Explain biosynthesis of & Catabolism of catecholamine.
- I] What is mean by narcotic antagonist. Draw structure & Therapeutic use of Naloxon & Levallorphan.

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